

Combinatorial Geometry

Labix

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1 Simplicies

1.1 Basic Definitions

Definition 1.1.1 (Affinely Independent)

Let $n, m \in \mathbb{N}$. We say that a set of points $\{v_0, \dots, v_m\} \subset \mathbb{R}^n$ is affinely independent if $v_1 - v_0, \dots, v_n - v_0$ are linearly independent.

Definition 1.1.2 (n -Simplexes)

Let $n \in \mathbb{N}$. Let $v_0, \dots, v_n \in \mathbb{R}^n$ be a set of affinely independent points. An n -simplex is the set of points

$$\Delta^n = \left\{ \sum_{k=0}^n t_k v_k \mid \sum_{k=0}^n t_k = 1 \text{ and } t_k \geq 0 \text{ for all } k = 0, \dots, n \right\}$$

We write $\Delta^n = [v_0, \dots, v_n]$ to indicate the spanning vectors. The standard n -simplex is just the n -simplex whose vertices are the standard basis vectors for $\mathbb{R}^{n+1}$.

Note that the vertices here are an ordered set so that an orientation is inherently defined. Realistically, the order of the vertices does not change the homology groups which we will define later.

Definition 1.1.3 (Properties of n -Simplexes)

- Let $\Delta^n = [v_0, \dots, v_n]$ be an n -simplex.
- The k th face of Δ^n is a the $n - 1$ simplex $\Delta_k^n = \Delta^n \cap \{x_k = 0\}$. We use $[v_0, \dots, \hat{v}_k, \dots, v_n]$ to indicate the k th $n - 1$ dimensional face
 - The boundary of Δ , written $\partial\Delta^n$ is the union of all its proper faces
 - The interior is defined to be $(\Delta^n)^\circ = \Delta^n \setminus \partial\Delta^n$

Definition 1.1.4 (The Vertex Set of a Simplex)

Let $S = [v_0, \dots, v_n] \subseteq \mathbb{R}^n$ be a simplex. Define the vertex set of the simplex to be

$$V(S) = \{v_0, \dots, v_n\}$$

It is easy to see that any face of an n -simplex is an $n - 1$ simplex in its own right.

Lemma 1.1.5 Any two k -simplexes where one in $\mathbb{R}^m$ and one in $\mathbb{R}^n$ are homeomorphic.

1.2 Barycentric Subdivision

Barycentric subdivisions is the main ingredient for proving one of the major results of singular homology. It is defined first through Δ -set, and then passed on to singular n -simplexes, which we will see in the next chapter.

Definition 1.2.1 (Barycenter) Let $v = [v_0, \dots, v_n]$ be an n -simplex. Define the barycenter of v to be the point

$$B(v) = \frac{1}{n+1} \sum_{i=0}^n v_i$$

In fact, we can find the barycenter inductively. Knowing the barycenter b_i on the i th face $[v_0, \dots, \hat{v}_i, \dots, v_n]$, let l_i be the line connecting b_i and v_i . Then b is the intersection of all these lines l_i .

Definition 1.2.2 (Barycentric Cone) Let v be an n -simplex. Let $w = [w_1, \dots, w_n] \subseteq v$ be an $(n - 1)$ -simplex in v . Define the barycentric cone of w to be the n -simplex

$$\mathcal{B}(w) = [B(v), w_1, \dots, w_n] \subset v$$

Intuitively, this means that given a face of Δ^n which is an $(n - 1)$ -simplex, its barycentric cone is the n -simplex constructed from the vertices of the face together with the barycenter.

Definition 1.2.3 (The Barycentric Map) Let v be an n -simplex. Define the barycentric map of v to be the map

$$\mathcal{B} : C_{n-1}(v) \rightarrow C_n(v)$$

defined on the basis by $\mathcal{B}([w_1, \dots, w_n]) = [B(v), w_1, \dots, w_n]$.

Definition 1.2.4 (Barycentric Subdivision) Let Δ^n be the standard n -simplex. Define inductively the barycentric subdivision

$$S^\Delta(\Delta^n) \in C_n(\Delta^n)$$

of Δ^n by the following recursive formula.

- When $n = 0$, $S^\Delta(\Delta^0) = \Delta^0$
- When $n > 0$, the barycentric subdivision of Δ^n is given by

$$S^\Delta(\Delta^n) = \sum_{i=0}^n (-1)^i \mathcal{B} S^\Delta(\partial_i \Delta^n)$$

To elicit a few examples, consider the case of $n = 1$. Then we have

$$\begin{aligned} S(\Delta^1) &= \mathcal{B} S(\partial \Delta^1) \\ &= \mathcal{B} S(\partial_0 \Delta^1) - \mathcal{B} S(\partial_1 \Delta^1) \\ &= \mathcal{B}[v_1] - \mathcal{B}[v_0] \\ &= [b, v_1] - [b, v_0] \end{aligned}$$

where $b = \frac{1}{2}(v_0 + v_1)$. Intuitively, we are breaking up the n -simplex Δ^n into tiny pieces of n -simplexes using the center of mass of each subsimplex in Δ^n .

Lemma 1.2.5 Let $\sigma = [w_1, \dots, w_n] \subseteq \Delta^n$ be an $(n-1)$ -simplex. Then the following are true.

- $\partial(\mathcal{B}(\sigma)) + \mathcal{B}(\partial\sigma) = \sigma$
- $\partial S(\Delta^n) = S(\partial \Delta^n)$

Proof We have that

$$\begin{aligned} \partial \mathcal{B}[w_1, \dots, w_n] &= \partial[b, w_1, \dots, w_n] \\ &= \sum_{i=0}^n (-1)^i \partial_i [b, w_1, \dots, w_n] \\ &= [w_1, \dots, w_n] - \mathcal{B}(\partial[w_1, \dots, w_n]) \end{aligned}$$

And thus the first identity is satisfied.

We prove the second item inductively. When $n = 0$, we have 0 on both sides. When $n > 0$, we have

$$\begin{aligned} \partial S \Delta^n &= \partial \mathcal{B}(S(\partial \Delta^n)) \\ &= \text{id}(S \partial \Delta^n) - \mathcal{B}(\partial(S \partial \Delta^n)) && \text{(First Identity)} \\ &= S \partial \Delta^n - \mathcal{B}(S \partial^2 \Delta^n) && \text{(Induction)} \\ &= S \partial \Delta^n \end{aligned}$$

■

Lemma 1.2.6 Let $v = [v_0, \dots, v_n]$ be a simplex. Let $[w_0, \dots, w_n]$ be a component in the barycentric subdivision of v . Then we have

$$\text{diam}([w_0, \dots, w_n]) \leq \frac{n}{n+1} \text{diam}([v_0, \dots, v_n])$$

Proof If $n = 0$, then the claim is true since $[w_0] = [v_0]$ has diameter 0. Assume that $n > 0$. We first prove the following subclaim: For every $x \in [x_0, \dots, x_n]$ an n -simplex, its maximum distance to points in the simplex is attained at a vertex x_i . Indeed if $y \in [x_0, \dots, x_n]$ with $\|x - y\|$ maximal, then we can write $y = \sum_{i=1}^n t_i x_i$ with $\sum_{i=1}^n t_i = 1$ and $t_i \geq 0$. Then we have that

$$\begin{aligned} \|x - y\| &= \left\| x - \sum_{i=1}^n t_i x_i \right\| \\ &= \left\| \sum_{i=1}^n t_i (x - x_i) \right\| \\ &\leq \sum_{i=1}^n t_i \|x - x_i\| \\ &\leq \max \|x - x_i\| \end{aligned}$$

with equality if y is one of the vertices with $\|x - x_i\|$ being maximal. By applying the above twice, we have that the diameter of $[w_0, \dots, w_n]$ is the length of the longest edge $[x, y]$ in $[w_0, \dots, w_n]$. Now there are two cases.

Case 1: None of x, y is the barycenter b of $[w_0, \dots, w_n]$. Then they must be the vertices of a simplex in the barycentric subdivision of one of the faces $[v_0, \dots, \hat{v}_i, \dots, v_n]$. By induction, we have that

$$\begin{aligned} \text{diam}([w_0, \dots, w_n]) &= \|x - y\| \\ &\leq \frac{n-1}{n} \text{diam}([v_0, \dots, \hat{v}_i, \dots, v_n]) \\ &\leq \frac{n}{n+1} \text{diam}([v_0, \dots, v_n]) \end{aligned}$$

and so we are done.

Case 2: Without loss of generality, $x = b$.

Then y lies on some face of $[v_0, \dots, v_n]$ and the claim above implies that we can take $y = v_i$ for some vertex v_i of that face. Let b_i be the barycenter of $[v_0, \dots, \hat{v}_i, \dots, v_n]$. Then we have that the barycenter of $[v_0, \dots, v_n]$ is

$$b = \frac{1}{n+1} \sum_{j=1}^n v_j = \frac{1}{n+1} v_i + \frac{n}{n+1} b_i$$

We thus have that

$$\begin{aligned} \text{diam}([w_0, \dots, w_n]) &= \|v_i - b\| \\ &\leq \frac{n}{n+1} \|v_i - b_i\| \\ &\leq \frac{n}{n+1} \text{diam}([v_0, \dots, v_n]) \end{aligned}$$

and so we conclude. ■

2 Simplicial and Delta Complexes

2.1 Simplicial Complexes

Definition 2.1.1 (Simplicial Complexes) Let $\mathcal{K}$ be a set of simplexes. We say that $\mathcal{K}$ is a simplicial complex if the following are true.

- For any $S \in \mathcal{K}$, every face $\partial_k S$ of S lies in $\mathcal{K}$.
- If S_1 and S_2 are two simplexes in $\mathcal{K}$, then either $S_1 \cap S_2$ is empty or $S_1 \cap S_2$ is a common face of both S_1 and S_2 .

Definition 2.1.2 (Abstract Simplicial Complex)

Let S be a set. We say that S is a simplicial complex if for all $\sigma \in S$ and $\tau \subseteq \sigma$, we have $\tau \in S$. Elements of S are called faces.

Example 2.1.3

The following are simplicial complexes.

- The n -polygon $\{\{i, i+1\} \mid 1 \leq i \leq n-1\} \cup \{\{1, n\}\}$.
- The n -simplex $\mathcal{P}(\{1, \dots, n+1\})$.
- The boundary of the n -simplex $\mathcal{P}(\{1, \dots, n+1\}) \setminus \{1, \dots, n+1\}$.

Example 2.1.4

The set $S = \mathcal{P}(\{1, 2, 3\}) \cup \mathcal{P}(\{1, 3, 4\}) \cup \mathcal{P}(\{2, 4\})$ is an abstract simplicial complex.

Definition 2.1.5 (Dimension)

Let S be a simplicial complex.

- Let $\sigma \in S$. Define the dimension of σ to be

$$\dim(\sigma) = |\sigma| - 1$$

- Define the dimension of S to be

$$\dim(S) = \max\{\dim(\sigma) \mid \sigma \in S\}$$

Definition 2.1.6 (Vertex Set)

Let S be a simplicial complex. Define the vertex set of S to be

$$V(S) = \bigcup_{\sigma \in S} V(\sigma)$$

Thus it is clear that $S \subseteq \mathcal{P}(V(S))$.

We mostly consider the case where $V(S)$ is finite. In this case, we may replace the underlying set by $\{1, \dots, n\}$ by a suitable $n \in \mathbb{N}$.

Definition 2.1.7 (Associated Abstract Simplicial Complex)

Let $\mathcal{K}$ be a simplicial complex. Define the associated abstract simplicial complex $A(\mathcal{K}) \subseteq \mathcal{P}(V(\mathcal{K}))$ as follows: For each simplex $\Delta \in \mathcal{K}$, we have $V(\Delta) \subseteq A(\mathcal{K})$.

Lemma 2.1.8

Let $\mathcal{K}$ be a simplicial complex. Then $A(\mathcal{K})$ is an abstract simplicial complex.

2.2 Maps between Simplicial Complexes

Definition 2.2.1 (Simplicial Maps)

Let S, T be simplicial complexes. Let $f : V(S) \rightarrow V(T)$ be a function. We say that f is a simplicial map if for each $\sigma \in S$, we have $f(\sigma) \in T$.

Lemma 2.2.2

Let S, T, U be simplicial complexes. Let $f : V(S) \rightarrow V(T)$ and $g : V(T) \rightarrow V(U)$ be simplicial maps. Then $g \circ f$ is a simplicial map.

Definition 2.2.3 (Isomorphism of Simplicial Complexes)

Let S, T be simplicial complexes. Let $f : V(S) \rightarrow V(T)$ be a function. We say that f is an isomorphism if there exists a simplicial map $g : V(T) \rightarrow V(S)$ such that

$$g \circ f = \text{id}_{V(S)} \quad \text{and} \quad f \circ g = \text{id}_{V(T)}$$

2.3 The Face Vector of a Simplicial Complex

Definition 2.3.1 (The Face Vector)

Let $\mathcal{K}$ be an abstract simplicial complex. Define the face vector $(f_0, \dots, f_{\dim(\mathcal{K})})$ of $\mathcal{K}$ by the formula

$$f_i = \text{The number of } i\text{-dimensional faces of } \mathcal{K}$$

for $0 \leq i \leq \dim(\mathcal{K})$. By convention, we set $f_{-1} = 1$.

Example 2.3.2

The following are true.

- The face vector of the n -polygon is given by (n, n) .
- The face vector of the n -simplex is given by $f_i = \binom{n+1}{i+1}$ for $0 \leq i \leq n$.
- The face vector of the boundary of the n -simplex is given by $f_i = \binom{n+1}{i+1}$ for $0 \leq i \leq n-1$.

Definition 2.3.3 (The h -Vector)

Let $\mathcal{K}$ be an abstract simplicial complex. Let $(f_0, \dots, f_{\dim(\mathcal{K})})$ be the face vector of $\mathcal{K}$. Define the h -vector $(h_0, \dots, h_{\dim(\mathcal{K})})$ of $\mathcal{K}$ by the formula

$$h_i = \sum_{k=0}^i (-1)^{i-k} \binom{n+1-k}{i-k} f_{k-1}$$

for $0 \leq i \leq \dim(\mathcal{K}) + 1$.

Example 2.3.4

The following are true.

- The h -vector of the n -polygon is given by $(1, -n^2, n(3-n)/2)$.
- The h -vector of the n -simplex is given by

$$h_i = \sum_{k=0}^i (-1)^{i-k} \binom{n+1}{i-k} \cdot (i-1) \cdots (i-k+1)$$

Lemma 2.3.5

Let S be an abstract simplicial complex on $\{1, \dots, n\}$. Let $(f_0, \dots, f_n)$ be the face vector of S . Let $(h_0, \dots, h_n)$ be the h -vector of S . Then the following are true.

- We have

$$\sum_{k=0}^{n+1} f_{k-1} (t-1)^{d-k} = \sum_{j=0}^{n+1} h_j t^{n+1-j}$$

for variable t .

- For $0 \leq i \leq n$, we have

$$f_i = \sum_{k=0}^{i+1} \binom{n+1}{i+1-k} h_k$$

2.4 Cubical Embeddings of Simplicial Complexes

Definition 2.4.1 (The Cubical Embedding)

Let $n \in \mathbb{N}$. Define the cubical embedding $i_c : \Delta^n \rightarrow [0, 1]^{n+1}$ as follows.

1. For $1 \leq k \leq n$, the barycenter of the k -dimensional face $B([e_{i_1}, \dots, e_{i_k}])$ is sent to $(u_1, \dots, u_{n+1}) \in [0, 1]^{n+1}$ where $u_j = 0$ if $j \in \{i_1, \dots, i_k\}$ and $u_j = 1$ otherwise.
2. Extend this into a piecewise linear embedding $i_c : \Delta^n \rightarrow [0, 1]^{n+1}$ as follows. For each simplex $[p_1, \dots, p_l]$ in the barycentric subdivision of Δ^n , $[i_c(p_1), \dots, i_c(p_l)]$ lies on a face of $[0, 1]^{n+1}$. Hence there exists an simplicial map sending $[p_1, \dots, p_l]$ to $[i_c(p_1), \dots, i_c(p_l)]$.

Lemma 2.4.2

Let $n \in \mathbb{N}$. Then $\partial[0, 1]^n$ is a simplicial complex.

Definition 2.4.3 (Cubical Embeddings of Abstract Simplicial Complexes)

Let $\mathcal{K}$ be an abstract simplicial complex. Define the cubical embedding of $\mathcal{K}$ to be the subspace

$$\text{cc}(\mathcal{K}) = i_c(|\mathcal{K}|)$$

3 Operations on Simplicial Complexes

3.1 Barycentric Subdivision

Recall that there is a notion of barycentric subdivision for simplicies. The definition easily extends to simplicial complexes, but not obvious for abstract simplicial complexes.

Definition 3.1.1 (Barycentric Subdivision)

Let $\mathcal{K}$ be a simplicial complex. Define the barycentric subdivision of $\mathcal{K}$ to be the abstract simplicial complex

$$\text{Bd}(\mathcal{K}) = \bigcup_{k=0}^{\infty} \{ \{v_0, \dots, v_k\} \subseteq \mathcal{P}(\mathcal{K}) \setminus \{\emptyset\} \mid v_0 \subset \dots \subset v_k \}$$

The vertex set of $\text{Bd}(\mathcal{K})$ is $\mathcal{K}$.

The idea is to add vertex for each simplex in the simplicial complex so that the underlying set of the barycentric subdivision is $\mathcal{K}$. Since $\mathcal{K}$ is a subset of $\mathcal{P}(\mathcal{K})$, elements of $\mathcal{K}$ are partially ordered by inclusion, each inclusion with the geometric meaning of faces of the simplex.

Lemma 3.1.2

Let $\mathcal{K}$ be a simplicial complex. Then we have

$$\dim(\text{Bd}(\mathcal{K})) = \dim(\mathcal{K})$$

Proposition 3.1.3

Let $\mathcal{K}$ be a simplicial complex. Then the following are true.

- For $0 \leq i \leq \dim(\text{Bd}(\mathcal{K}))$, the face vector for $\text{Bd}(\mathcal{K})$ is given by

$$f_i(\text{Bd}(\mathcal{K})) = \sum_{j=i}^{\dim(\mathcal{K})} b_{ij} f_j(\mathcal{K})$$

where

$$b_{ij} = \sum_{k=0}^i (-1)^k \binom{i+1}{k} (i-k+1)^{j+1}$$

- The matrix $(b_{ij}) \in M_{\dim(\mathcal{K})+1}(\mathbb{Z})$ is upper triangular and invertible.
- The face vector for $\text{Bd}(\mathcal{K})$ is given by matrix multiplication:

$$f(\text{Bd}(\mathcal{K})) = (b_{ij}) f(\mathcal{K})$$

Proposition 3.1.4

Let $\mathcal{K}$ be a simplicial complex. Then the following are true.

- For $0 \leq i \leq \dim(\text{Bd}(\mathcal{K}))$, the face vector for $\text{Bd}(\mathcal{K})$ is given by

$$h_i(\text{Bd}(\mathcal{K})) = \sum_{j=0}^{\dim(\mathcal{K})+1} d_{ij} h_j(\mathcal{K})$$

where

$$d_{ij} = \sum_{k=0}^i (-1)^k \binom{\dim(\mathcal{K})+2}{k} (i-k)^j (i-k+1)^{\dim(\mathcal{K})+1-j}$$

- The matrix $(d_{ij}) \in M_{\dim(\mathcal{K})+2}(\mathbb{Z})$ is invertible.
- The h -vector for $\text{Bd}(\mathcal{K})$ is given by matrix multiplication:

$$h(\text{Bd}(\mathcal{K})) = (d_{ij}) h(\mathcal{K})$$

3.2 Links and Stars

Definition 3.2.1 (The Link of a Face)

Let S be a simplicial complex. Let $\sigma \in S$. Define the link of σ to be

$$\text{lk}(\sigma) = \{\tau \in S \mid \sigma \cup \tau \in S \text{ and } \sigma \cap \tau = \emptyset\}$$

Definition 3.2.2 (The Star of a Face)

Let S be a simplicial complex. Let $\sigma \in S$. Define the star of σ to be

$$\text{st}(\sigma) = \{\tau \in S \mid \sigma \cup \tau \in S\}$$

Define the boundary of the star of σ to be

$$\partial \text{st}(\sigma) = \{\tau \in S \mid \sigma \cup \tau \in S \text{ and } \sigma \not\subseteq \tau\}$$

Lemma 3.2.3

Let S be a simplicial complex. Let $p \in V(S)$. Then we have

$$\text{lk}(p) = \partial \text{st}(p)$$

3.3 The Join of Simplicial Complexes

Definition 3.3.1 (The Join of Two Simplicial Complexes)

Let $\mathcal{K}_1, \mathcal{K}_2$ be simplicial complexes. Define the join of $\mathcal{K}_1$ and $\mathcal{K}_2$ to be

$$\mathcal{K}_1 * \mathcal{K}_2 = \{I_1 \amalg I_2 \mid I_1 \in \mathcal{K}_1, I_2 \in \mathcal{K}_2\}$$

Example 3.3.2

We have $\Delta^n * \Delta^m = \Delta^{n+m-1}$.

3.4 The Connected Sum of Simplicial Complexes

4 Realizing Simplicial Complexes in Euclidean Space

4.1 Geometric Realization

Definition 4.1.1 (Geometric Realization with Respect to Function)

Let S be an abstract simplicial complex. Let $n \in \mathbb{N}$. Let $\phi : V(S) \rightarrow \mathbb{R}^n$ be a function. Define the geometric realization of S with respect to ϕ to be the space

$$|S|_\phi = \bigcup_{\sigma \in S} [\phi(s) \mid s \in \sigma]$$

Definition 4.1.2 (Affine Embedding)

Let S be an abstract simplicial complex. Let $\phi : V(S) \rightarrow \mathbb{R}^n$ be a function. We say that $|S|_\phi$ is an affine embedding if ϕ is injective.

Proposition 4.1.3

Let S be an abstract simplicial complex. Let $\phi : V(S) \rightarrow \mathbb{R}^n$ and $\psi : V(S) \rightarrow \mathbb{R}^m$ be affine embeddings. Then $|S|_\phi$ and $|S|_\psi$ are homeomorphic.

Since any two affine embeddings give homeomorphic geometric realizations, we often drop the underlying injective function and assume that a choice has been chosen.

Definition 4.1.4 (Geometric Realization of a Simplicial Map)

Let S, T be simplicial complexes. Let $\phi_1 : V(S) \rightarrow \mathbb{R}^n$ and $\phi_2 : V(T) \rightarrow \mathbb{R}^m$ be affine embeddings. Let $f : S \rightarrow T$ be a simplicial map. Define the geometric realization of f to be the function

$$|f| : |S|_{\phi_1} \rightarrow |T|_{\phi_2}$$

given as follow. For each $x \in |S|$, there exists $\lambda_v \in \mathbb{R}$ such that $x = \sum_{v \in T_1} \lambda_v \phi_1(v)$ and $\sum_{v \in V(S)} \lambda_v = 1$. Then define

$$|f|(x) = \sum_{v \in V(S)} \lambda_v \psi(f(v))$$

Proposition 4.1.5

Let S_1, S_2 be simplicial complexes. Let $f : S \rightarrow T$ be a simplicial map. Then the following are true.

- Let $\sigma \in S$. Then we have

$$|f|(|\sigma|) = |f(\sigma)|$$

- Let $\sigma \in S$. Then $|f|_{|\sigma|}$ is affine linear.

Lemma 4.1.6

Let S, T, U be simplicial complexes. Let $f : S \rightarrow T, g : T \rightarrow U$ be simplicial maps. Then we have

$$|g \circ f| = |g| \circ |f|$$

Thus geometric realization is a functor from a pair (S, ϕ) where S is a simplicial complex on some set T and ϕ is an affine embedding, to its geometric realization with respect to the embedding.

Definition 4.1.7 (Standard Geometric Realization of Abstract Simplicial Complexes)

Let $n \in \mathbb{N}^+$. Let S be an abstract simplicial complex on $\{1, \dots, n\}$. Define the standard geometric realization of S to be the space

$$|S| = \bigcup_{I \in S} \Delta^I$$

where Δ^I is the $|I|$ -simplex with vertices e_i for $i \in I$.

By definition, the geometric realization $|S|$ is a subcomplex of the standard n -simplex $[e_1, \dots, e_{n+1}] \subseteq \mathbb{R}^{n+1}$.

Example 4.1.8

Let $n \in \mathbb{N}$. Consider the set $S = \mathcal{P}(\{1, \dots, n+1\})$. Then $|S|$ is the standard n -simplex.

4.2 Properties of Geometric Realization

Lemma 4.2.1

Let S be a simplicial complex. Then $|S|$ and $|\text{Bd}(S)|$ are homeomorphic.

Lemma 4.2.2

Let S be a simplicial complex. Let $p \in V(S)$. Then we have

$$|\text{st}(p)| = \bigcup_{F \in S, p \in F} |F|$$

Lemma 4.2.3

Let $\mathcal{K}_1, \mathcal{K}_2$ be simplicial complexes. Then we have

$$|\mathcal{K}_1 * \mathcal{K}_2| = |\mathcal{K}_1| * |\mathcal{K}_2|$$

TBR (cones and suspensions are not yet introduced).

Corollary 4.2.4

Let $\mathcal{K}$ be a simplicial complex. Then the following are true.

- $|\Delta^0 * \mathcal{K}| = C|\mathcal{K}|$ (the cone of the space).
- $|\partial \Delta^1 * \mathcal{K}| = \Sigma|\mathcal{K}|$.

4.3 Simplicial Approximation

Definition 4.3.1 (Simplicial Approximations)

Let S, T be simplicial complexes. Let $f : |S| \rightarrow |T|$ be a continuous map. A simplicial approximation of f is a simplicial map $F : V(S) \rightarrow V(T)$ such that $|F|$ is homotopic to f .

Theorem 4.3.2 (The Simplicial Approximation Theorem)

Let S, T be simplicial complexes. Let $f : |S| \rightarrow |T|$ be a continuous map. Then there exists $n \in \mathbb{N}^+$ and a simplicial approximation $F : \text{Bd}^n(S) \rightarrow T$ of f .

Proposition 4.3.3

Let X be a CW complex. Then there exists a simplicial complex S such that $|S|$ is homotopic to X .

4.4 Triangulated Spaces

Definition 4.4.1 (Triangulated Spaces)

Let X be a space. We say that X is triangulated if there is a simplicial complex K such that $|K| \cong X$.

5 Delta Complexes and Simplicial Homology

5.1 Delta Complexes

Definition 5.1.1 (Δ -Complexes)

A Δ -complex $(S_\bullet, d_\bullet)$ consists of the following data.

- For each $n \in \mathbb{N}$, a set S_n of n -simplices.
- For each $n \in \mathbb{N}$ and $0 \leq i \leq n$, a map $d_i^n : S_n \rightarrow S_{n-1}$ such that

$$d_i^{n-1} \circ d_j^n = d_{j-1}^{n-1} \circ d_i^n$$

called the face relation is satisfied whenever $i < j$.

In particular, d_i sends an n -simplex to its i th face, which is an $n - 1$ simplex. This means that for $s = [v_1, \dots, v_n] \in S_n$, $d_i(s) = [v_1, \dots, \hat{v}_i, \dots, v_n]$.

Proposition 5.1.2 Every simplicial complex is a Δ -complex.

Definition 5.1.3 (Geometric Realization of Δ -Complexes)

Let $S = (S_\bullet, d_\bullet)$ be a Δ -set. The geometric realization of S is a topological space X that is built up inductively as follows:

- The 0-skeleton X^0 is a discrete set with points in S_0
- Given the $n - 1$ skeleton X^{n-1} and S_{n-1} . Now define

$$X^n = \left(X^{n-1} \cup \coprod_{\alpha \in I_n} \Delta_\alpha^n \right) / \sim$$

where $\sim$ is the equivalence relation $\Delta_\beta^{n-1} \sim \partial_k \Delta_\alpha^n$ given from the face maps $d_\bullet$. (Intuitively, each face of the n -simplex in S_n gets identified with a $n - 1$ simplex in X^{n-1})

- Define $X = \bigcup_n X^n$. The minimal n for which $X = X^n$ is called the dimension of n

We also write X as $|S|$ to indicate the Δ -set.

Definition 5.1.4 (Map of Delta-Complexes)

Let $S = (S_\bullet, d_\bullet)$ and $S' = (S'_\bullet, d'_\bullet)$ be two Δ -sets. A map of Δ -sets $f : S \rightarrow S'$ is a family of maps $f_n : S_n \rightarrow S'_n$ for each $n \in \mathbb{N}$ such that for all $0 \leq i \leq n$, we have that

$$d'_i \circ f_n = f_{n-1} \circ d_i : S_n \rightarrow S'_{n-1}$$

5.2 Delta Complex Structures

Δ -complexes act much nicer than CW complex because they are combinatorial. Indeed the attaching maps d_j^n given by the Δ -set are combinatorial in nature: we just need to define which element in S_{n-1} each face of the simplex gets mapped to. In other words, $d_k^n : S_n \rightarrow S_{n-1}$ maps each $\Delta \in S_n$ to its k th face in S_{n-1} .

Theorem 5.2.1 The geometric realization of a Δ -complex is a CW complex

However, most of the topological spaces are CW complexes rather than Δ -complexes. Indeed the notion of CW complexes are less restrictive on the attaching maps, while that of Δ -complexes are predetermined the face maps: one has to glue them so that each face of the n -simplexes is identified as an $n - 1$ simplex in the existing skeleton.

Definition 5.2.2 (Delta-complex Structure) Let X be a topological space. A Δ -complex structure on X is a Δ -complex S together with a homeomorphism $|S| \cong X$.

The Δ -complex structure is not unique. For example, one can have multiple ways of even defining a circle, depending the number of points.

5.3 Simplicial Homology

The main goal is now to associate to every Δ -complex an abelian group. This abelian group will serve as an invariant of the Δ -complex. This is a two step process. To every Δ -complex S , we associate a chain complex and then a collection of homology groups

$$S \mapsto (\Delta_\bullet(S), \partial_\bullet) \mapsto H_\bullet(S)$$

with both steps being functorial in the sense that it respects associativity and identity when given a map of Δ -complexes.

We begin by treating the case of Δ -sets.

Definition 5.3.1 (Simplicial n -Chains) Let $S = (S_\bullet, d_\bullet)$ be a Δ -set. Define the group of simplicial n -chains on S to be free group

$$\Delta_n(S) = \mathbb{Z}\langle \Delta_k^n \mid \Delta_k^n \in S_n \rangle$$

on the set of n -simplexes. An n chain is then of the form

$$\sum_k m_k \Delta_k^n$$

for $m_k \in \mathbb{Z}$ and $\Delta_k^n \in S_n$.

All the simplicial n -chains are related by a formula called the boundary operator.

Definition 5.3.2 (Boundary Operator) Let S be a Δ -set. Define the boundary operator $\partial_n : \Delta_n(S) \rightarrow \Delta_{n-1}(S)$ by

$$\partial_n(s) = \sum_{k=0}^n (-1)^k d_i^n(s)$$

Proposition 5.3.3 The family of abelian groups $\Delta_n(S)$ of a Δ -set S and the boundary operator ∂ forms a chain complex

$$(\Delta_\bullet(S), \partial_\bullet)$$

In particular, $\partial_n \circ \partial_{n+1} = 0$ for all $n \in \mathbb{N}$ where ∂_n is the boundary operator above.

Proof Let $s \in \Delta_{n+1}(X)$. Then we have that

$$(\partial_n \circ \partial_{n+1})(s) = \partial_n \left(\sum_{j=0}^{n+1} (-1)^j d_j^{n+1}(s) \right) = \sum_{i=0}^n \sum_{j=0}^{n+1} (-1)^{i+j} d_i^n(d_j^{n+1}(s))$$

Fix a pair $0 \leq i < j \leq n+1$. By definition 2.1.5, $A = (-1)^{i+j} d_i^n(d_j^{n+1}(s))$ and $B = (-1)^{i+j} d_{j-1}^n(d_i^{n+1}(s))$ cancel out. Moreover every summand is of the form A or B and not both so that the sum vanishes. In other words, we have that

$$\begin{aligned} \sum_{i=0}^n \sum_{j=0}^{n+1} (-1)^{i+j} d_i^n(d_j^{n+1}(s)) &= \sum_{0 \leq i < j \leq n+1} (-1)^{i+j} d_i^n(d_j^{n+1}(s)) + \sum_{0 \leq j < i \leq n} (-1)^{i+j} d_i^n(d_j^{n+1}(s)) \\ &= \sum_{0 \leq i < j \leq n+1} (-1)^{i+j} d_{j-1}^n(d_i^{n+1}(s)) + \sum_{0 \leq j < i \leq n} (-1)^{i+j} d_i^n(d_j^{n+1}(s)) \\ &= \sum_{0 \leq i \leq j \leq n} (-1)^{i+j-1} d_j^n(d_i^{n+1}(s)) + \sum_{0 \leq j \leq i \leq n} (-1)^{i+j} d_i^n(d_j^{n+1}(s)) \\ &= 0 \end{aligned}$$

We conclude that $(\Delta_\bullet(S), \partial_\bullet)$ forms a chain complex. ■

Recalling from section 1, we can now define the homology groups of the chain complex.

Definition 5.3.4 (The Simplicial Homology Groups) Let S be a Δ -set and $(\Delta_\bullet(S), \partial_\bullet)$ the chain complex of S .

- Define the group of n -cycles to be $Z_n(S) = \ker(\partial_n)$
- Define the group of n -boundaries to be $B_n(S) = \text{im}(\partial_{n+1})$
- Define the n th simplicial homology group to be the quotient

$$H_n^\Delta(S) = \frac{Z_n(S)}{B_n(S)} = \frac{\ker(\partial_n)}{\text{im}(\partial_{n+1})} = H_n(\Delta_\bullet(S))$$

We will see different forms of chain complexes in the rest of the notes. In fact, we have seen from section 1 that to every chain complex we can associate a sequence of homology groups. In our case of simplicial homology, there is a very topological way of understanding the homology groups. But this is not necessarily true for general homology groups built out of arbitrary chain complexes.

Definition 5.3.5 (Simplicial Homology Groups of a Geometric Realization) Let X be a topological space with a Δ -complex structure $|S| \cong X$, define its n th simplicial homology group to be

$$H_n^\Delta(X) = H_n^\Delta(S)$$

By definition, $H_n^\Delta(X)$ would be nonzero exactly when the cycles in X , quotiented out with boundary of the faces is exactly the n dimensional holes. This is because cycles in X does not necessarily capture holes as there may be some faces within the cycles. Therefore we have to quotient out the cycles that encapture faces.

Since $H_n(C_\bullet)$ in general just means the homology groups of the chain complex $C_\bullet$, to distinguish between different homologies arising from different chain complexes, we will give special names for specific homology groups. In this instance, the simplicial homology of X will be denoted $H_n^\Delta(X)$ instead of just $H_n(X)$. This is also to distinguish between other types of homology on X , such as singular homology.

We have seen that a map between two chain complexes induces a map between the homology groups. We can now define a map between two Δ -sets so that they induce a map in chain complexes and in turn, a map in simplicial homology groups.

Lemma 5.3.6 A map of Δ -sets $f_\bullet : S \rightarrow S'$ induces a chain map $f_\bullet : \Delta_\bullet(S) \rightarrow \Delta_\bullet(S')$.

Proof Define $f_n : \Delta_n(S) \rightarrow \Delta_n(S')$ on each generator $s \in S_n$ by $s \mapsto f_n(s)$ and extend it linearly on the free group. Then f_n is naturally a group homomorphism. We now verify that f_n together with the boundary operators satisfy the required commutativity. For $s \in S_n$, we have that

$$\begin{aligned} \partial'_n(f_n(s)) &= \sum_{k=0}^n (-1)^k d_k^n(f_n(s)) \\ &= \sum_{k=0}^n (-1)^k f_{n-1}(d_k(s)) \\ &= f_{n-1} \left(\sum_{k=0}^n (-1)^k d_k(s) \right) \\ &= f_{n-1}(\partial_n(s)) \end{aligned}$$

and so we conclude. ■

Combining with the induced map in homology, we thus have that given a map of Δ -sets, it induces a map in homology.